

<u>Artificial Intelligence</u> <u>Course Instructor</u> Ms. Mahzaib Younas		
Time allowed = 45 min	Quiz 3	Total Marks = 35

BCS Section A

_____	_____	_____
Roll No	Name	Signature

Question No 01: For each question statement below, choose the correct option.[6]

1. What is the main characteristic of local search algorithms? a) They operate on multiple nodes at a time b) They operate using single node & move to neighbors c) They systematically can explore all of the possible paths. d) They store all the visited nodes in the memory	2. What is the key advantage of local search algorithms? a) They guarantee in finding the optimal solution. b) They use little memory and are efficient in large state spaces. c) They always avoid getting stuck in local optima. d) They do not require an evaluation function
3. In hill climbing, what happens when the algorithm reaches a local maximum? a) It restarts automatically from a new state b) It always finds a way to reach the global maximum c) It stops since no neighboring state has a higher value d) It moves randomly to another state	4. Which of the following is NOT a type of hill climbing? a) Stochastic hill climbing b) Steepest ascent hill climbing c) First-choice hill climbing d) Depth-first hill climbing
5. What is the role of temperature in simulated annealing? a) It determines the probability of accepting worse solutions b) It controls the number of neighbors a state can have c) It determines the final solution directly d) It has no impact on the algorithm	6. What are the main cons of hill-climbing search? a) Terminates at local optimum & Does not find optimum solution b) Terminates at global optimum & Does not find optimum solution c) Does not find optimum solution & Fail to find a solution d) Fail to find a solution

Q2. Answer the following

[3 Marks]

Statement	True/False	Justification
Local search algorithms always find the optimal solution.	False	It always stuck on the local maxima.
Simulated annealing is an improved version of hill climbing that allows bad moves.	True	Yes, both work with only one neighbor value.
In local beam search, increasing the value of k allows more states to be explored simultaneously.	True	K number of neighbors to explore the k values
Random-restart hill climbing helps to escape local optima by starting from different initial states.	True	Random initial state can be better start points to find global maxima
In simulated annealing, if temperature (T) is never lowered, the algorithm will always stay in a high-energy state.	True	T = possible bad movies, reduce T to stop accepting bad move.

Question No 03: Consider an **8-Queens Problem** where the heuristic function is:
 $h = \text{Number of pairs of queens attacking each other}$

[6 Marks]

You apply **hill climbing** but the algorithm gets stuck at a **plateau**.

Questions:

a) What does a **plateau** mean in this context?

A plateau is a region in the state space where neighboring states have the same heuristic value—no reduce the number of pairs of queens attacking each other.

b) Give an example of how a **random sideways move** can help escape a plateau.

If moving a queen results in no change in attacking pairs (i.e., h -value remains constant), a sideways move can explore new possibilities and potentially lead to an improvement later.

c) How does **simulated annealing** behave differently from hill climbing in this case?

Simulated annealing will not stop when reached the local optima because of T . It will allow some queens to take bad move that can increase no of queens attacking but can lead to better solution.

Question No 04: Give the name of the algorithm that results from each of the following special cases: [1.5 x 4 = 6]

1. Simulated annealing with a temperature $T = 0$ also behaves identically to a hill climbing search.

Hill Climb Search, because $T=0$ means it will not accept any bad move.

2. Local beam search with one initial state and no limit on the number of states retained.

Breadth first search because it explore the whole search state.

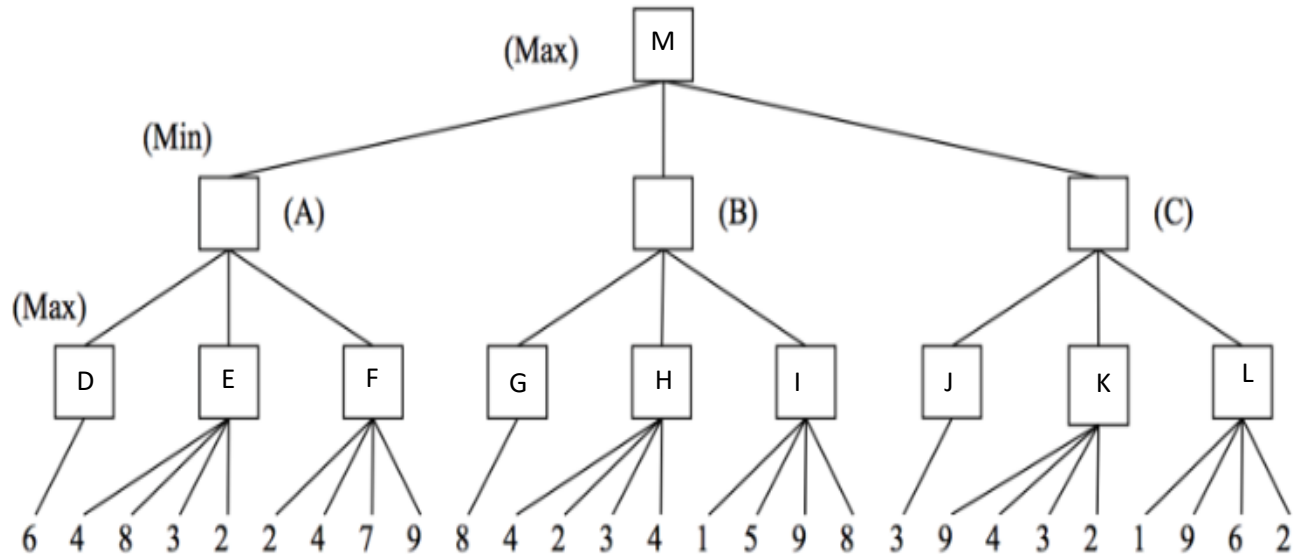
3. Simulated annealing with $T = \infty$ at all times

Random walk, it will never stop.

4. Local beam search with $k = 1$.

Hill Climb search because Local beam only explore the one neighbor.

Question No 05: Consider a Game tree below and assuming both rational agents are playing the game with MAX agent is having a first turn. Use Alpha beta pruning to find optimal path for MAX agent. As we know ordering affects the efficiency of Pruning, you are required to prune this tree in both left to right and right to left order. [7 + 7 = 14 Marks]

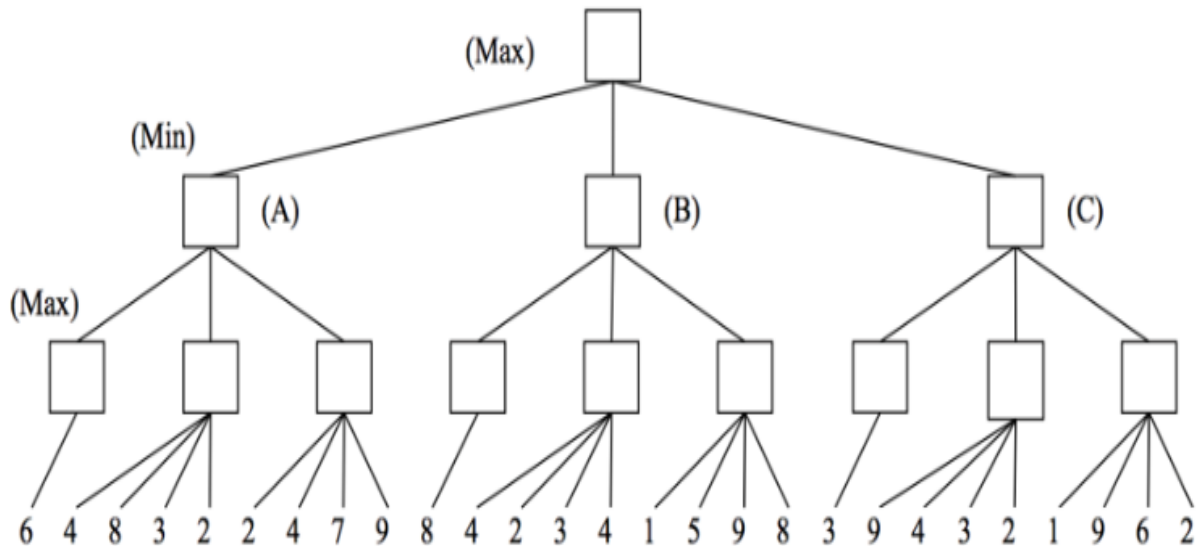


1. write the new value if any changes occur later on each level

	A	B	C	D	E	F	G	H	I	J	K	L	M
alpha	$-\infty$	$-\infty, 6$	$-\infty, 6$	$-\infty, 6$	$-\infty, 8$	$-\infty, 7$	$-\infty, 8$	$-\infty, 4$		$-\infty, 3$			$-\infty, 6$
Beta	$+\infty, 6$	$+\infty, 8, 4$	$+\infty, 3$	$+\infty$	$+\infty, 6$	$+\infty, 6$	$+\infty$	$+\infty, 8$		$+\infty$			$+\infty$

2. Write names of all nodes in left to right order which will get pruned

E = 3, 2
F = 9
I = all nodes
K = all nodes
L = all nodes



1. write the new value if any changes occur later on each level

	A	B	C	D	E	F	G	H	I	J	K	L	M
alpha	$-\infty, 4$	$-\infty, 3$	$-\infty$	$-\infty, 6$	$-\infty, 8$	$-\infty, 9$	$-\infty, 8$	$-\infty, 4$	$-\infty, 9$	$-\infty, 3$	$-\infty, 9$	$-\infty$	$-\infty, 3, 4, 6$
Beta	$+\infty, 9$	$+\infty, 9, 4$	$+\infty, 3$	$+\infty, 9$	$+\infty$	$+\infty$	$+\infty, 9$	$+\infty, 9$	$+\infty$	$+\infty, 9$	$+\infty, 9$	$+\infty, 9$	$+\infty$

2. Write names of all nodes in right to left order which will get pruned

No nodes.

3. Write complete optimal paths for MAX Agent in both L->R and R->L Order, starting from node A

4. How Ordering affects the efficiency of pruning? Which order is more efficient?